

Forecasting PGA Tour Finishing Position Using Time Series of Strokes-Gained Metrics

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Abstract

This study examines which strokes-gained category contributes most to finishing position on the PGA tour. We focus on one specific player: Bryson DeChambeau. Multiple models are estimated, including different models for the different strokes-gained metrics. Comparing models using Akaike Information Criterion (AIC) shows that putting performance exhibits persistence and nonlinear behavior, and finishing position is mostly determined by strokes-gained in its respective tournament, not previous ones. Overall, the results show that off-the-tee and approach are the greatest drivers of finishing position.

Introduction

“Strokes-gained” is a measure of players’ shot quality in different categories in relation to other PGA Tour players. It can be broken up into four main categories: off-the-tee, approach, around-the-green, and putting, making it easy for analysts to determine which aspect of the game contribute most to scoring in relation to the field. While strokes-gained is known to be related to success, it is less known how each category relates to finishing position.

The goal of this paper is to determine which of these categories contributes most to finishing position when performance is treated as a time series. This allows results to reflect persistence across tournaments. In golf, players tend to remain at a similar performance level for extended periods of time, so this method is relevant.

Bryson DeChambeau is the focus of this study. His resume is interesting and useful because he is an esteemed off-the-tee and approach player, with above-average around-the-green statistics and noticeably variable putting.

Literature Review

Previous research shows that strokes-gained is a strong indicator of PGA Tour results, with approach shots carrying the most weight. Shots from outside 100 yards account for roughly two-thirds of the total score among PGA Tour professionals (Broadie, 2012). Researchers have also shown that strokes-gained metrics can provide insight into future outcomes (Drappi & Ting Keh, 2018).

Recent research shows that putting is very similar to white noise, and that tee-to-green (all strokes-gained categories excluding putting) are more stable over time (Brill and Wyner 2025). Shot and skill level models suggests that separating underlying skill from noise improves performance evaluation (Drappi and Ting Keh 2018).

Data

This study uses a Kaggle dataset of PGA Tour results covering the years 2015 through 2022. For each tournament and player, it records finishing position and several strokes-gained measures, including performance off-the-tee, on approach shots, around-the-green, and in putting.

Missing values are handled on a model-by-model basis to preserve as much information as possible without distorting the data. A summary of the dataset and variable structure is provided in Appendix Figure A1.

Methodology

First, linear regression models establish a baseline of how each strokes-gained category impacts finishing position. Polynomial, spline, and generalized additive models allow for nonlinear relationships. Time-series structure is examined using ARIMAX and VAR specifications to assess whether past information improves model fit beyond current performance measures.

For univariate strokes-gained putting models, AIC prefers long-memory specifications such as FARIMA and TAR over simpler ARIMA and exponential smoothing models. This shows that putting performance is persistent but nonlinear. AIC comparisons for these models are reported in Appendix Table A4.

Hidden Markov Models are analyzed because they provide a description of hot and cold putting regimes (although they are not the lowest AIC univariate models).

Model Selection

Univariate analysis of strokes-gained putting clearly shows nonlinear behaviour. Models with long memory and threshold effects fit the data better, indicating that the short-term noise does not fully describe putting and that putting gradually evolves over time. This explains why, although putting is noisy, hot and cold regimes still remain meaningful.

Hidden Models are analyzed despite not having the lowest AIC of the univariate models. Two and three-state models provide a framework for putting streakiness, highlighting hot and

cold regimes. AIC comparisons for the HMM specifications and state summaries are reported in Appendix Table A2, with regime paths shown in Appendix Figure A4.

Regression models outperform time-series models in predicting finishing position. Generalized additive models (GAM) have the lowest AIC of the finishing-position models. In a close second is linear and polynomial regressions. This suggests that allowing nonlinear relationships between strokes-gained and finishing position explains most of the variation across tournaments without needing to model past outcomes. Full AIC comparisons for finishing-position models are reported in Appendix Table A1.

ARIMAX and VAR models yield high AIC values. This indicates that analyzing finishing position as a time series does not improve model fit when strokes-gained components are included. These models suggest that allowing outcomes to depend on past tournament results adds little explanatory power after controlling for strokes-gained.

Results

Regression results indicate that finishing position is primarily driven by ball striking (off-the-tee and approach) rather than short game (around-the-green and putting). Baseline regressions show that off-the-tee has the greatest coefficient, while around-the-green has the least impact. Full coefficient estimates are reported in Appendix Table A3.

Advanced regression models lead to similar conclusions. Polynomial, spline, and GAM models allow the relationship between strokes-gained and finishing position to vary across performance levels. While this shows some nonlinearity, they do not produce different findings, so ball striking remains the most impactful. GAM smooths are shown in Appendix Figure A3.

Treating finishing position as time series does not improve model fit once strokes-gained components are added. ARIMAX and VAR models underperform regression when comparing

their AIC. This suggest that finishing position is mostly determined by the current tournaments' strokes-gained statistics rather than previous events. ARIMAX output is shown in Appendix Figure A2, with AIC comparisons reported in Appendix Table A1.

In contrast, univariate analysis of putting shows a strong time-series behaviour. FARIMA and TAR models fit much better than ARIMA, indicating that overall putting performance gradually evolves over time despite short-term noise. These results motivate the use of regime-based models and are summarized in Appendix Table A3.

Hidden Markov Models identify hot and cold putting regimes. Transition probabilities suggest that once Bryson DeChambeau enters a particular regime, he tends to remain there for multiple tournaments. Regime paths over time are shown in Appendix Figure A4, with state-level summary statistics reported in Appendix Table A2.

Clustering further supports the impact of ball-striking. His better finishes cluster around better off-the-tee and approach statistics, while putting and around-the-green vary within clusters. This suggests that ball striking distinguishes between Bryson's good and bad finishes, as illustrated in Appendix Figure A3.

Clustering applied to the full sample of players confirms these findings. Averages indicate that clusters are primarily determined by off-the-tee and approach, which putting and around the green have overlap amongst clusters. Cluster-level strokes-gained summaries are reported in Appendix Table A5.

Discussion

Clearly, finishing position is mostly explained by overall strokes-gained performance, with ball-striking playing the dominant role. Off-the-tee and approach shots contribute most to finishing position, while around-the-green has less impact and putting is the most volatile.

Putting adds context rather than having a large impact. It is not pure noise, and the univariate results indicate persistence and regime behavior. However, these effects do not outweigh ball-striking in the long run.

Bryson DeChambeau's tournament results clearly cluster around off-the-tee and approach components, and his good finishing positions are driven mostly by ball striking rather than extended putting streaks.

Conclusion

This paper analyzes the relationship between strokes-gained metrics and finishing position using various models. While strokes-gained putting exhibits persistent and nonlinear behavior, finishing position is best explained off-the-tee and approach shots.

These findings support the previous literature that tournament results depend more on consistently good ball striking than putting streaks. Future research could extend this framework to additional players, but results would likely be similar.

Appendix

Appendix Figure A1

Tournament-Level Data Overview

tournament id	date	pos	sg_putt	sg_ott	sg_app	sg_arg	sg_total
<dbl>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2250	2015-06-14	45	0.31	0.29	0.72	-0.51	0.81
2726	2016-03-16	2	NaN	NaN	NaN	NaN	NaN
2493	2016-04-10	21	NaN	NaN	NaN	NaN	NaN
2494	2016-04-17	4	-0.01	0.25	1.38	1.02	2.65
2499	2016-06-05	38	-0.74	0.12	-0.02	1.43	0.79
2501	2016-06-19	15	NaN	NaN	NaN	NaN	NaN

6 rows

tournament id	date	pos	sg_putt	sg_ott	sg_app
Min. : 2250	Min. :2015-06-14	Min. : 1.00	Min. : -1.2200	Min. : -1.1400	Min. : -1.790
1st Qu. : 3748	1st Qu. :2018-01-14	1st Qu. : 6.25	1st Qu. : -0.1600	1st Qu. : 0.2950	1st Qu. : -0.165
Median :401056518	Median :2019-02-19	Median :19.00	Median : 0.3600	Median : 0.6900	Median : 0.290
Mean :281647956	Mean :2019-03-03	Mean :22.53	Mean : 0.4859	Mean : 0.7048	Mean : 0.324
3rd Qu. :401155473	3rd Qu. :2020-07-26	3rd Qu. :35.75	3rd Qu. : 1.1050	3rd Qu. : 1.0100	3rd Qu. : 0.990
Max. :401366873	Max. :2022-01-09	Max. :66.00	Max. : 2.5000	Max. : 2.0100	Max. : 1.950
		NA's :19	NA's :19	NA's :19	NA's :19
sg_arg	sg_total	cluster_kmeans	cluster_hc	cluster_timepoint	
Min. : -2.00000	Min. : -1.640	Min. :1.000	Min. :1.00	Min. :1.000	
1st Qu. : -0.37000	1st Qu. : 0.565	1st Qu. :1.000	1st Qu. :1.00	1st Qu. :2.000	
Median : 0.03000	Median : 1.420	Median :2.000	Median :2.00	Median :2.000	
Mean : 0.03587	Mean : 1.551	Mean :1.947	Mean :1.76	Mean :2.307	
3rd Qu. : 0.41000	3rd Qu. : 2.650	3rd Qu. :3.000	3rd Qu. :2.50	3rd Qu. :3.000	
Max. : 1.43000	Max. : 5.570	Max. :3.000	Max. :3.00	Max. :3.000	
NA's :19	NA's :19	NA's :19	NA's :19	NA's :19	

Appendix Table A1

AIC Comparison Across Finishing Position Models

	Model	AIC
	<chr>	<dbl>
ARIMAX_sg_diff1_with_sg_total	ARIMAX_sg_diff1_with_sg_total	140.4346
HMM_2_state_SG_putt	HMM_2_state_SG_putt	198.4277
HMM_3_state_SG_putt	HMM_3_state_SG_putt	202.0922
GAM_pos_on_SG	GAM_pos_on_SG	563.4867
Polynomial_regression_SG2	Polynomial_regression_SG2	567.0524
Linear_regression_pos_on_SG	Linear_regression_pos_on_SG	568.6156
Spline_regression_SG_bs4	Spline_regression_SG_bs4	575.4537
VAR_pos_and_SG	VAR_pos_and_SG	1186.9177

8 rows

Appendix Figure A2

ARIMAX Model Output

```

Series: y
Regression with ARIMA(4,0,0) errors

Coefficients:
      ar1      ar2      ar3      ar4      xreg
    -0.7515 -0.3917 -0.4843 -0.2788  0.0106
s.e.    0.1335  0.1617  0.1505  0.1305  0.0244

sigma^2 = 0.375: log likelihood = -64.22
AIC=140.43  AICc=141.69  BIC=154.26

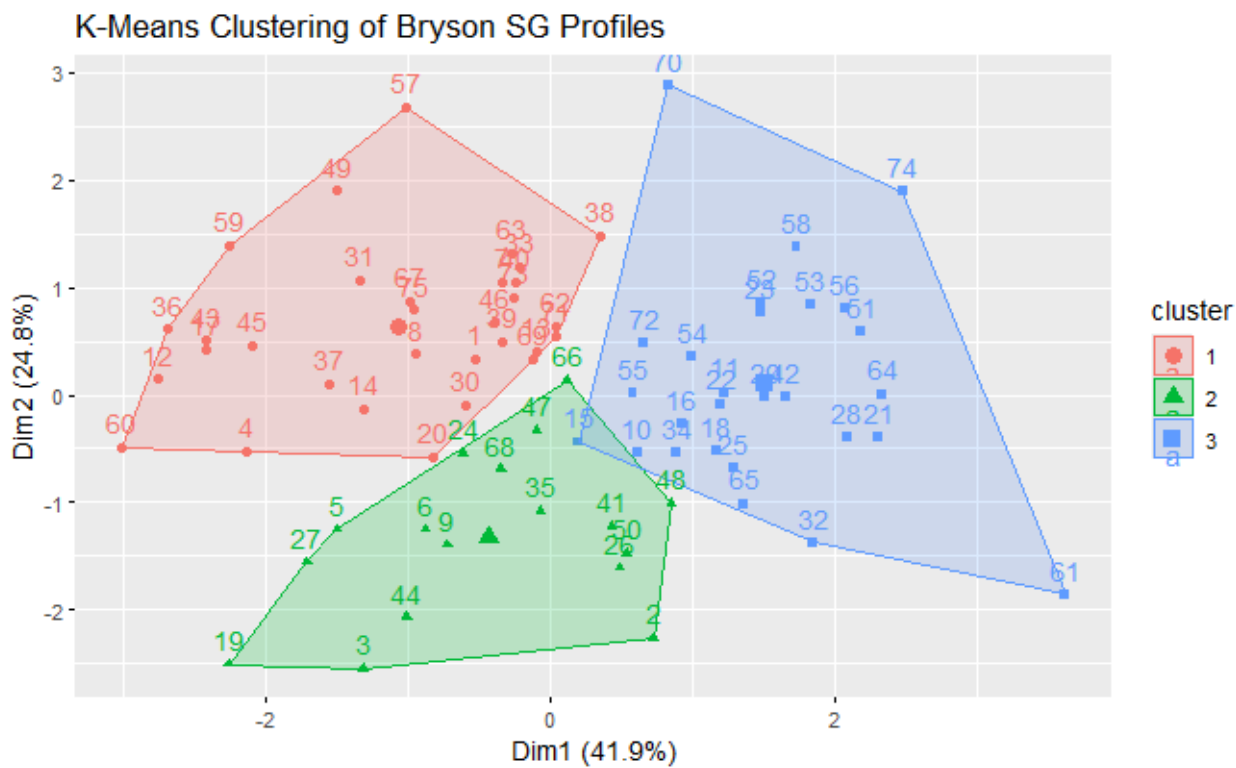
Training set error measures:
              ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 0.04609559 0.6567168 0.5367246 -33.72607 250.8583 0.4077701 -0.03868529

z test of coefficients:
      Estimate Std. Error z value Pr(>|z|)
ar1  -0.751527   0.133525 -5.6284 1.819e-08 ***
ar2  -0.391702   0.161686 -2.4226 0.015409 *
ar3  -0.484322   0.150531 -3.2174 0.001294 **
ar4  -0.278775   0.130548 -2.1354 0.032727 *
xreg   0.010573   0.024350  0.4342 0.664129
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

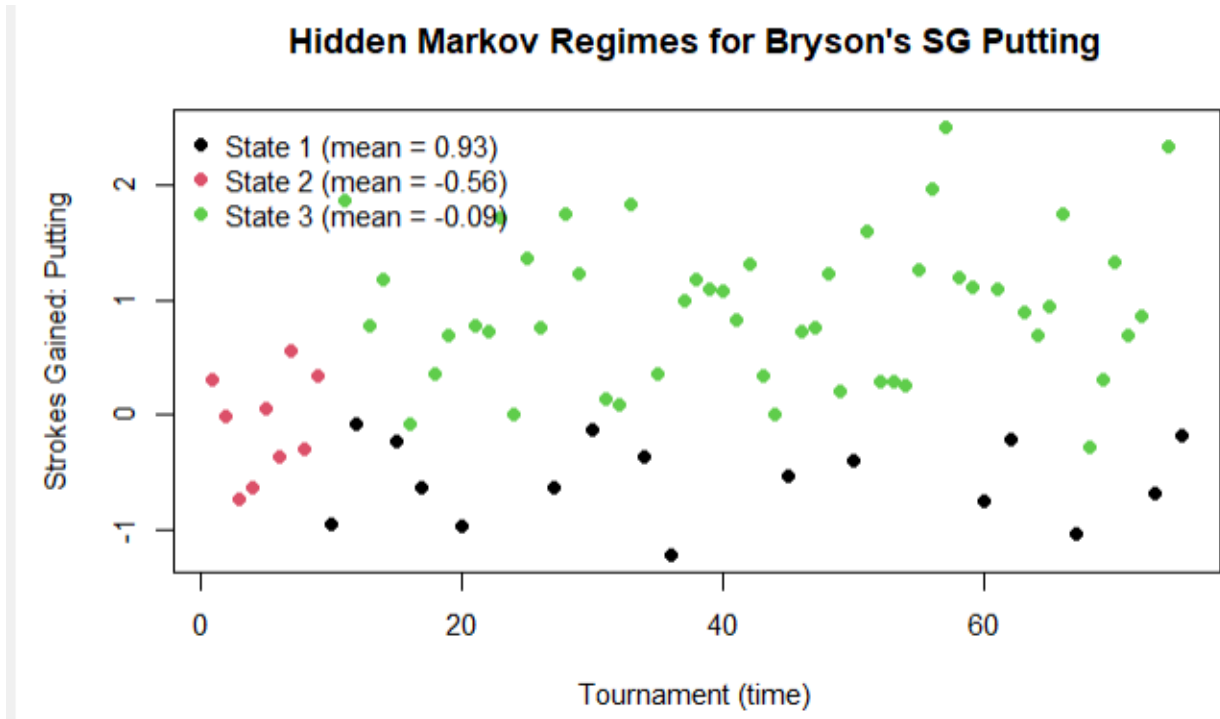
Appendix Figure A3

K-Means Clustering of Tournament Performance Profiles



Appendix Figure A4

Hidden Markov Model Putting Regimes Over Time



Appendix Table A2

Hidden Markov Model State Summary

State	Mean_SG_Putt	SD_SG_Putt
1	0.925600	0.6361785
2	-0.564375	0.3526465
3	-0.090000	0.4484139

	To State 1	To State 2	To State 3
From State 1	7.140625e-01	2.859375e-01	6.237942e-33
From State 2	9.999996e-01	3.790189e-07	5.401677e-51
From State 3	1.187837e-27	1.356745e-01	8.643255e-01

Appendix Table A3

Regression

term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
(Intercept)	37.614782	1.991629	18.886438	3.465778e-29
sg_putt	-10.263050	1.413779	-7.259302	4.196302e-10
sg_ott	-11.775426	2.145411	-5.488657	6.095052e-07
sg_app	-10.178250	1.493450	-6.815261	2.707651e-09
sg_arg	-8.677568	2.076973	-4.177987	8.357509e-05

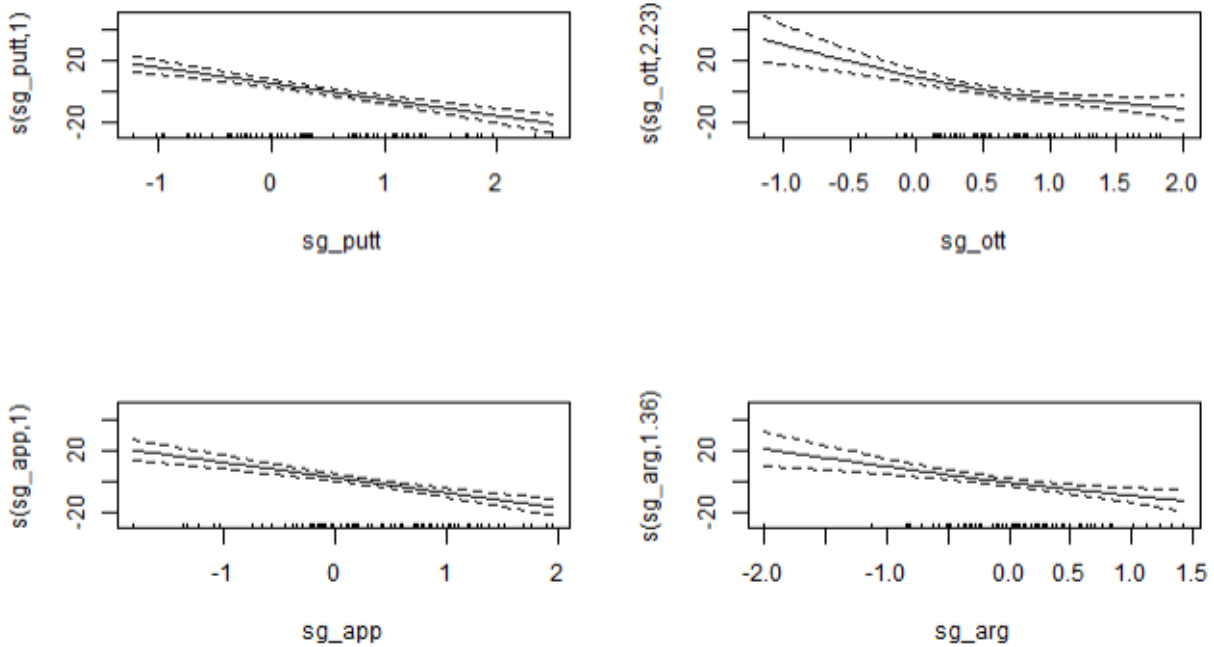
Appendix Table A4

AIC Comparison Across Univariate SG Putting Models

group <chr>	model <chr>	AIC <dbl>
SG Putting univariate	FARIMA (arfima)	-59.948988
SG Putting univariate	TAR m=3	-46.711130
SG Putting univariate	TAR m=2	-42.914118
SG Putting univariate	TAR m=2 delay=1	-37.550830
SG Putting univariate	EGARCH(1,1) t	2.727801
SG Putting univariate	GARCH(1,1) normal	2.787360
SG Putting univariate	GARCH(1,1) t	2.815474
ARIMAX	ARIMAX on sg_diff1 with xreg=sg_total	140.434607
SG Putting univariate	Drift (Arima 0,0,0 drift)	146.606160
SG Putting univariate	SARIMA (auto.arima on sg_diff1)	150.703126
ARIMAX	ARIMAX on sg_putt levels with xreg=sg_total	165.291913
HMM (SG Putting regimes)	HMM 2-state (Gaussian)	198.427717
SG Putting univariate	Naive (Arima 0,0,0 no mean)	200.444496
HMM (SG Putting regimes)	HMM 3-state (Gaussian)	202.092178
SG Putting univariate	Mean (Arima 0,0,0 mean)	202.442180
SG Putting univariate	ETS	262.256888
Finishing position models	GAM: pos ~ s(sg_putt)+s(sg_ott)+s(sg_app)+s(sg_arg)	563.486666
Finishing position models	Polynomial regression (degree 2)	567.052429
Finishing position models	Linear regression: pos ~ sg_putt + sg_ott + sg_app + sg_arg	568.615602
Finishing position models	Spline regression (bs df=4)	575.453711
Finishing position models	VAR (pos, sg_putt, sg_ott, sg_app, sg_arg)	1186.917700

Appendix Figure A5

GAM Smooth Terms for strokes-gained Components



Appendix Table A5

Cluster-Level Mean strokes-gained Components (All Players)

	cluster <int>	sg_putt <dbl>	sg_ott <dbl>	sg_app <dbl>	sg_arg <dbl>	sg_total <dbl>
1	1	-0.24072619	-0.63579607	-0.6450631	-0.16539056	-1.6805841
2	2	-0.28964162	0.02437879	-0.2533068	-0.20672832	-0.7215462
3	3	0.01399079	-0.26156595	-0.1965700	0.07425362	-0.3658051
4	4	-0.11512374	0.23551092	0.2358297	0.02272542	0.3847141

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